

AI-HDL 2026: Design Phase 3 Documentation

Phase Title: Security Evaluation and Threat Mitigation

Duration: March 12 – March 26, 2026

Objective: To take your Optimized DP#2 design and make it secure. Focus on Security Assessment, Countermeasure Implementation, and Validation.

1. Weekly Milestone Journey

This phase is structured as a 2-week sprint to ensure a stable foundation for the following optimization phases.

Week 1 (Mar 12-18): Security Assessment & Implementing Security Features

- Create a "Security Review Document" using current Threat models like CIA, STRIDE, and DREAD.
 - **The CIA Triad:** The goal is to ensure data remains secret (Confidentiality), is never altered by a hacker (Integrity), and the hardware is always ready to work when needed (Availability).
 - **STRIDE:** This is a checklist used to imagine how a hacker might attack us, from "Spoofing" a user's identity to "Elevating" their privileges to gain total control over the chip.
 - **DREAD:** This is a scoring system (1–10) that helps us decide which security bugs are the most dangerous and need to be fixed immediately versus those that are low risk.
- Use LLMs to identify threats and vulnerabilities in your codebase.
- List potential **Common Weakness Enumerations** (common hardware bugs like buffer overflows or insecure debug ports) relevant to your DP2 design.
- Develop a "Mitigation Plan" for every hole you found, decide on a "fix" (e.g., adding encryption, locking a port, or masking signals).
- Use LLMs to counter identified threats and vulnerabilities.

Week 2 (Mar 19-25): Validation, Hardening & Documentation

- Implement at least one major security countermeasure. (e.g., Add privilege-level checks, data input authentication, or logic to randomize power consumption).
- The "Proof of Security" Test: How do you know it works? Try to "break" your own design. If you added a lock, show that a "hacker" input now gets blocked.
- Run your original tests to make sure the new security features didn't break normal functionality or the original purpose of the chip.
- Analyze the PPA impact. How much did your security feature "cost" in area or timing?

- Write your final DP3 Security Report and tag your secure RTL for submission.

2. Deliverables & Submission Guidelines

Teams must submit their work via a private GitHub repository, granting access to assigned mentors and judges.

Required for DP3 Submission:

- **Secure RTL:** Your modified, hardened, and functionally correct Verilog code.
- **Security Evaluation Report:**
 - ☐ Your vulnerability assessment from Week 1.
 - ☐ A detailed explanation of the countermeasures you implemented.
 - ☐ Validation results (how you proved the fix works).
 - ☐ PPA overhead analysis (the "cost" of your security feature).
- **GitHub Tag:** A final commit tagged as "DP3-Submission".

3. Grading & Advancement Criteria

DP3 accounts for **25% of the total cumulative score**.

Category	Weight	Criteria
Security Feature Implementation	30%	Meaningful, well-implemented countermeasures.
Validation & Hardening	25%	Comprehensive security testing; prove the fix works.
Security Assessment	20%	Thorough vulnerability review; well-structured plan.
Documentation & Reporting	15%	Clear DP3 report; detailed explanation of mitigations & trade-offs.
Design Stability	10%	Functional correctness is preserved; all regression tests are clean.

4. Useful links

- AI-HDL website link: <https://csm.arizona.edu/AIHDL>
- AI-HDL Github link: <https://github.com/prismlabarizona/AIHDL-2026>
- Design phase 3 example: <https://youtu.be/bSISaX4ymXg>
- Design phase 2 example: <https://www.youtube.com/watch?v=fUBtewf05rE>
- Design phase 1 example: https://www.youtube.com/watch?v=M20AJiT_ETE&t=33s
- Webinar 3 – Base design Implementation:
<https://www.youtube.com/watch?v=rOaLiDxNm74>
- RISC-V code repo: <https://github.com/TinyTapeout/ttsky25a-tinyQV>

- Peripheral template repo: <https://github.com/prismlabarizona/AIHDL-2026/tree/master/tinyqv-full-peripheral-template>
- Yosys Installation method: <https://www.youtube.com/watch?v=XRexMO1B6s8&pp=2AYE>
- OpenLane Installation method: <https://www.youtube.com/watch?v=tFaTHGgpslA&pp=2AYB>

Important Date: Milestone Review 3 (MR#3) is scheduled for **March 26, 2026**. Ensure your mentor check-ins are completed weekly to receive advisory feedback.